

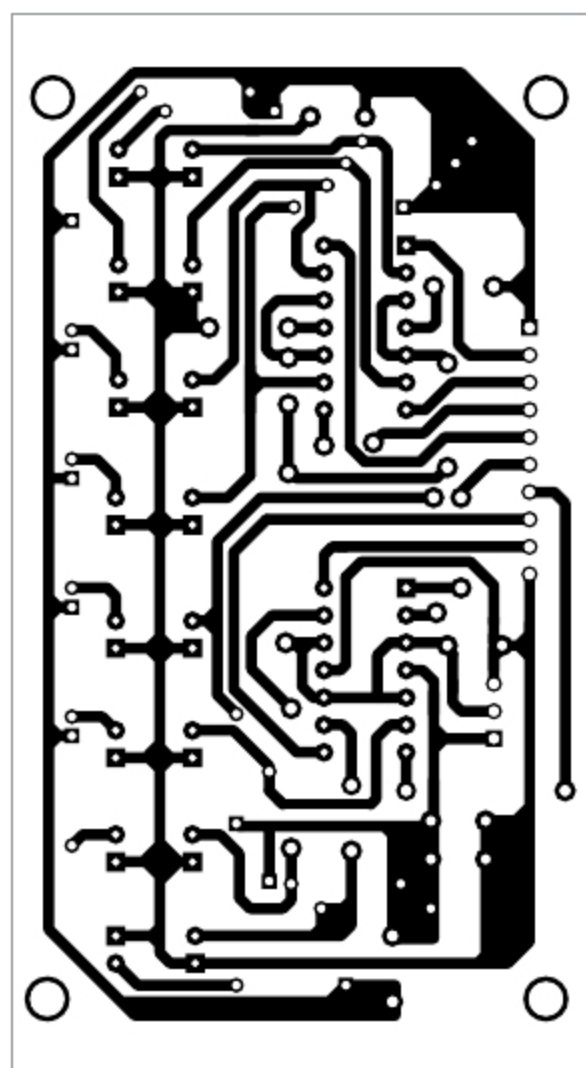


Fig. 6: Actual-size PCB layout for sensor unit

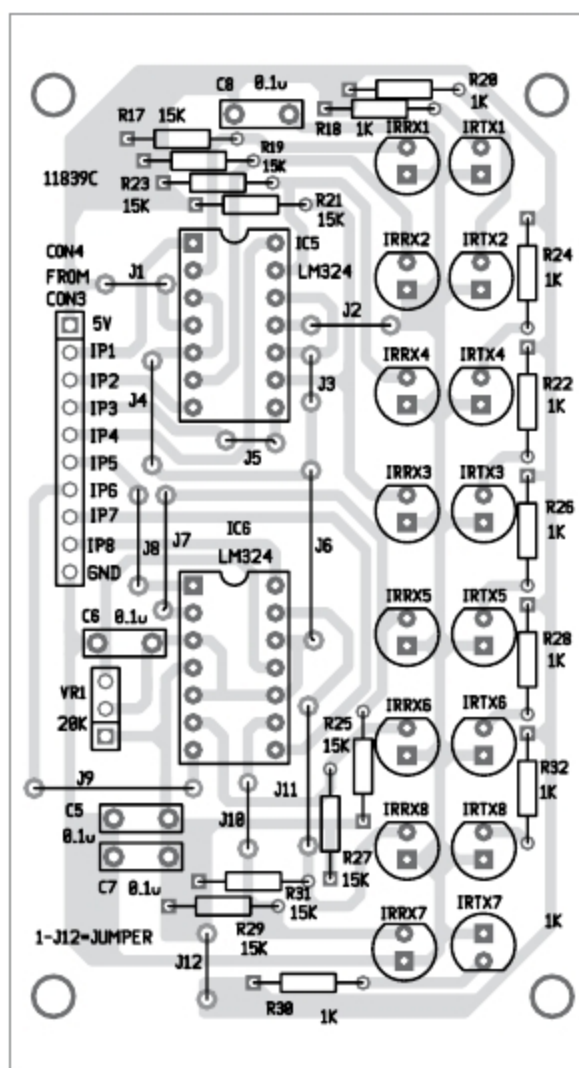


Fig. 7: Components layout of the PCB for Fig. 6

PARTS LIST

Semiconductors:

- IC1 - ATmega16 microcontroller
- IC2 - 7805 5V voltage regulator
- IC3 - 7806 6V voltage regulator
- IC4 - L293D motor driver
- IC5, IC6 - LM324 operational amplifier
- LED1-LED3 - 5mm LED
- IRTX1-IRTX8 - IR transmitting LED
- IRRX1-IRRX8 - IR receiving LED
- OLED - 2.4cm (0.96") OLED display

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R15, R16 - 2.2-kilo-ohm
- R2-R5 - 10-kilo-ohm
- R6-R14 - 4.7-kilo-ohm
- R17, R19, R21, R23, R25, R27, R29, R31 - 15-kilo-ohm
- R18, R20, R22, R24, R26, R28, R30, R32 - 1-kilo-ohm
- VR1 - 20-kilo-ohm potentiometer

Capacitors:

- C1, C3 - 100 μ F, 25V electrolytic
- C2, C4 - 10 μ F, 16V electrolytic
- C5-C8 - 0.1 μ F ceramic disk

Miscellaneous:

- BATT.1 - 11.1V LiPo battery
- CON1 - 4-pin connector
- CON2 - 6-pin connector
- CON3, CON4 - 10-pin male and female connector
- S1-S4 - Tactile switch
- BAR1 - 10-LED bar array
- M1, M2 - 6V DC motor
- J1 - Shorting jumper with 2-pin male Berg connector
- 2-pin terminal connector (3 number)

Running and stopping

Press button S3 to Run or Stop the line follower when not in configuration mode. Press reset button S4 anytime to stop and restart the operation again.

The current mode Run/Stop is displayed on the top right corner of the OLED. Initially, in Stop mode, set some value (say 8.0) for KP while keeping the values of KI and KD at zero. Make a thick black curved track on the floor and keep the line follower robot on top of the track. Press button S3 to run the line follower robot. It should follow the curved line drawn on the floor. To adjust the accuracy of the robot movement, change the values of KD and KI.

The searchLine () function in the code is called when no line is found. This function is independent of the PID algorithm and may be changed as per the individual's requirement, if any. **EFY**

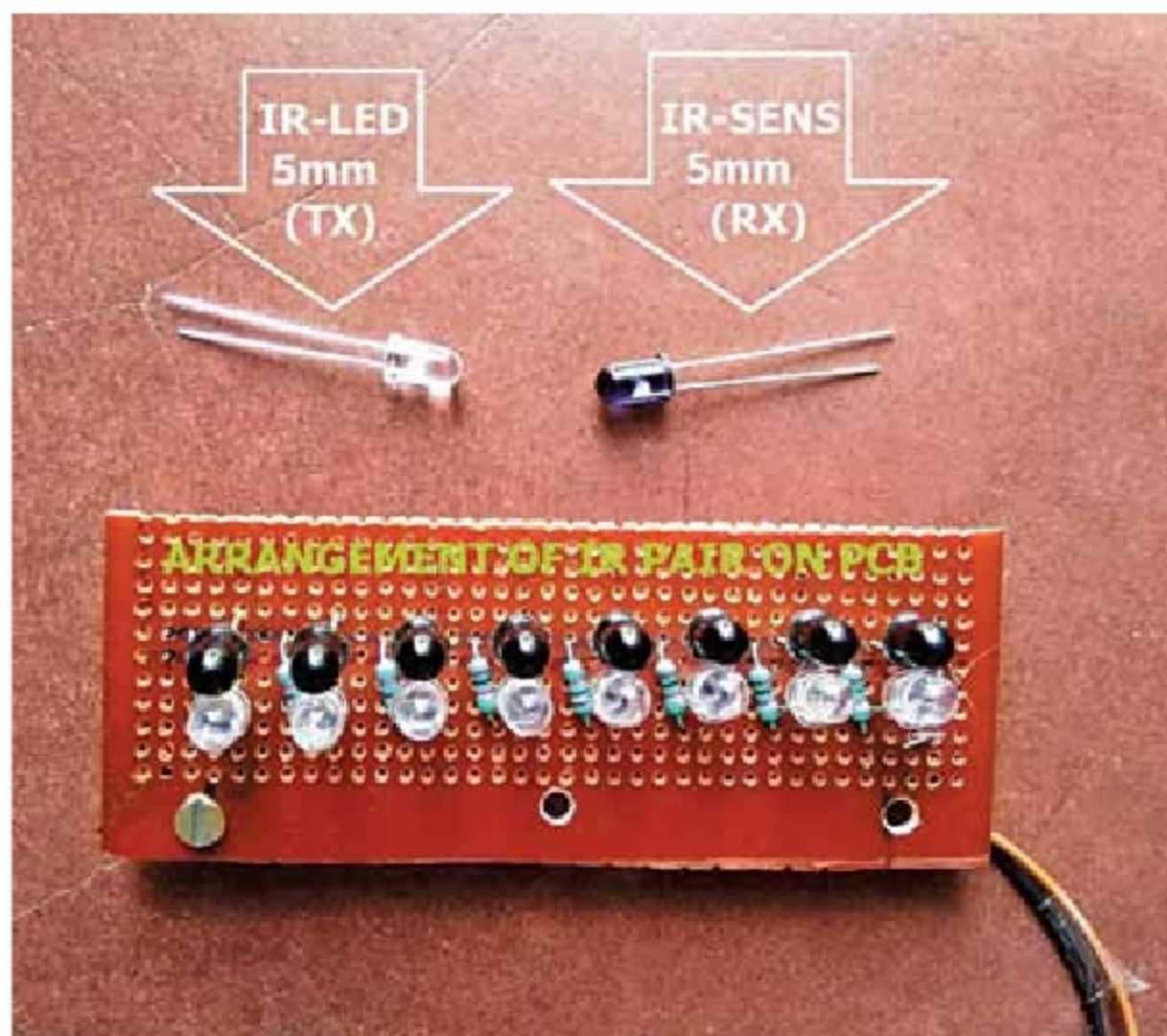


Fig. 8: Author's sensor unit assembly

DELAY) is the delay between readings of line data by the IR sensors. You can change it to say Delay = 50

5. Search loop (displayed as

SEARCH) is the number of times the program searches for a line before coming to stop mode. Change it to say Search = 45



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